A Prebiotic Chemistry "Time Machine"

Emergent Synthesis of Life's Molecular Toolkit from Bare Atoms via Free-Market Agent-Based Modeling

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Abstract. We present a computational "time machine" that replays prebiotic chemistry: starting from bare C, H, O, and N atoms, a free-market agent-based model grounded in physics — Bond Dissociation Energies (BDE) and Universal Force Field (UFF) strain evaluation — discovers thermodynamically favorable reaction pathways and **spontaneously produces life's molecular toolkit**: all 12 feasible amino acids, **all 5 nucleobases** (the letters of RNA and DNA), the **complete formose sugar chain** from formaldehyde to ribose to glucose, **Krebs cycle intermediates** (pyruvate, succinate, malate), and the first **tripeptide** (Gly-Gly-Gly). The simulation runs on a standard laptop in under 5 minutes, discovering up to 260 independent synthesis routes for a single amino acid. No target molecules are specified; all products emerge purely from thermodynamic drive. Amino acid formulas appear **14× earlier** than random chance would predict, confirming the thermodynamic inevitability hypothesis. Beyond origins-of-life research, the framework offers a rapid exploration tool for **novel synthesis pathway design** in modern chemistry.

1. Compressing the Chemical Clock

In H.G. Wells' *The Time Machine* (1895), the protagonist builds a device to witness evolution unfold across millennia. We propose an analogous device for chemistry: imagine peering into a virtual micelle — a tiny compartment filled with C, H, O, and N atoms — and pressing fast-forward through hundreds of millions of years. Like a time-lapse movie, you watch atoms collide, bond, and build increasingly complex molecules until, in just minutes of simulation time, the building blocks of life emerge from the primordial soup.

The question of how amino acids arose on prebiotic Earth has been studied through four independent approaches:

Laboratory experiments Miller & Urey (1953) passed electric sparks through $CH_4/NH_3/H_2O/H_2$ and found amino acids after days. Simple, but slow and hard to instrument at the molecular level.	**Meteorite analysis** The Murchison meteorite (1969) contains the same amino acids, confirming abiotic synthesis occurs in space — no biology required.
Quantum chemistry DFT and ab initio methods compute individual reaction pathways at supercomputer cost. Weeks per pathway; exploring the full combinatorial space is impractical.	**Thermodynamic theory** Higgs & Pudritz (2009) showed that free energy rankings predict amino acid abundance. Elegant, but reveals products, not pathways.

We introduce a **fundamentally different approach**: a Darwinian Agent-Based Model (D-ABM) implementation of a Free Market economic system, where chemistry is enacted by virtual workers — think of tiny dwarfs or daemons — who run workshops, trade molecules with their neighbors, and discover new reactions by trial and error. Each workshop transforms input molecules into products, guided only by energy: if the product is more stable (lower energy), the reaction succeeds. Supply and demand drive which workshops thrive. The system naturally discovers thermodynamically favorable pathways — and, remarkably, converges on the same amino acids, nucleobases, sugars, and metabolic intermediates that nature selected as the molecular toolkit of life.

2. The Model: Chemistry as a Free Market

2.1 The Economic–Chemical Mapping

The simulation transposes concepts from the DEPLOYERS framework — a Darwinian ABM originally developed for self-organizing economic systems (Jaraiz, 2020) and recently validated for macroeconomic forecasting with real-world FIGARO data (Jaraiz, 2026) — to chemistry:

Economic Concept	Chemical Analog	Example
Workers / Dwarfs	Universal agents with domain-specific behavioral rules	900 agents (one per atom)
Goods	Atoms, Molecules	C, H, O, N, CH_4, H_2O, glycine
Producers / Firms	Chemical reactions	$CH_3NO_2 + CH_4 \rightarrow C_2H_5NO_2$
Trade	Molecular exchange	Neighbor gives CH_4, receives C_2H_6
Consumer demand (GFCF)	Thermodynamic drive	Workers prefer larger, more complex molecules
Startup / Innovation	Reaction discovery	UFF force field validates new bond formation

2.2 Energy Evaluation

A reaction proceeds only if $\Delta E < 0$ (exothermic):

$$\Delta E = E_{strain}(\text{product}) - \text{BDE}(\text{new bond})$$

where BDE is the Bond Dissociation Energy (e.g., C–C = 346, C=O = 799, C=N = 615 kJ/mol) and E_{strain} is the Universal Force Field (UFF) strain energy from RDKit 3D geometry optimization.

For molecules larger than 30 atoms, UFF noise exceeds the signal, so we use BDE only — or evaluate a local fragment (atoms within 2 bonds of the reaction site, ~8 atoms) for meaningful local strain discrimination.

Thermodynamic equilibrium with kinetic filtering. The model focuses on thermodynamic favourability ($\Delta E < 0$) but includes an Evans-Polanyi kinetic barrier: $E_a = E_0 + \alpha \cdot \Delta E$ (E_0=50 kJ/mol, α=0.3), with Boltzmann acceptance $P = \exp(-E_a/kT)$. Strongly exothermic reactions ($\Delta E < -167$ kJ/mol) always proceed; weakly exothermic reactions are probabilistically suppressed.

2.3 Bond Types

Bond	BDE (kJ/mol)	When it forms
C–C	346	Default condensation bond
C–O	358	Condensation with oxygen
C–N	305	Condensation with nitrogen
C=O	**799**	Heteroatom double bond (anti-cumulene rule)
C=N	**615**	Heteroatom double bond
O–H, N–H	459, 386	Primary bonding
Ring closure	varies	BFS shortest path + UFF validation

2.4 The Workshop Manager's Day

Each simulation step, every agent manages its workshop and acts as a consumer. As **consumers**, workers generate demand via GFCF (Gross Fixed Capital Formation). As **workshop managers**, they handle production using only intermediate consumption (IC — input molecules):

1. **Discover** — Try combining an inventory molecule with a neighbor's molecule. If $\Delta E < 0$, a new recipe is born.
2. **Buy goods (GFCF)** — As consumers, workers prefer larger, more complex molecules, generating demand that propagates through the supply chain.
3. **Manage workshop** — Close unprofitable workshops; protect the last producer of each product. These "essential-service" producers stabilize the system, analogous to state-owned enterprises.
4. **Try startup** — Open a new workshop if idle and demand exists.
5. **Run workshop** — Produce using known recipes from IC inputs. Try to grow by combining the product with a neighbor's goods.

3. Results: The Time Machine in Action

3.1 Setup

Reactor: 200 C + 500 H + 100 O + 100 N = 900 atoms | **Mode:** Full discovery (blind) | **Temperature:** 288 K | **Neighborhood:** 30 agents | **Discovery budget:** 10 UFF evals/step

3.2 Amino Acid Emergence

Within 500 steps (~5 minutes on a laptop), **12 of the 12 feasible amino acid formulas** emerge spontaneously:

#	Amino Acid	Formula	First Step	Routes	Events	Status
1	**Threonine**	$C_4H_9NO_3$	**9**	106	108	*First!*
2	**Valine**	$C_5H_{11}NO_2$	**10**	132	133	
3	**Leucine**	$C_6H_{13}NO_2$	**11**	91	92	
4	**Alanine**	$C_3H_7NO_2$	**12**	172	173	
5	**Aspartate**	$C_4H_7NO_4$	**17**	31	32	
6	**Glycine**	$C_2H_5NO_2$	**30**	240	241	*Most routes*
7	**Glutamate**	$C_5H_9NO_4$	**32**	49	50	
8	**Asparagine**	$C_4H_8N_2O_3$	**59**	108	109	
9	**Glutamine**	$C_5H_{10}N_2O_3$	**80**	98	99	
10	**Serine**	$C_3H_7NO_3$	**113**	172	173	
11	**Histidine**	$C_6H_9N_3O_2$	**317**	14	16	*Late (3N needed)*
12	**Proline**	$C_5H_9NO_2$	**400**	3	3	*Ring amino acid*

3.3 The "14× Early" Effect

Key finding: By step 50, only 213 molecular formulas have been discovered — yet **7 of 12 amino acid formulas** are already among them (3.3%). Over the full 2,000-step run, 5,056 formulas are discovered, of which amino acids represent only 0.24%.

Amino acids appear **14× earlier than random chance** would predict. They occupy energy minima in C,H,O,N combinatorial space.

Step	Total Species	Amino Acids Found	AA / Species
10	49	**2**	4.1%
50	213	**7**	3.3%
100	326	**9**	2.8%
500	789	**12**	1.5%
2,000	1,308	**12**	0.9%

3.4 The Eight That Don't Form — and Why

Amino Acid	Formula	Missing Feature	Evolutionary Note
Cysteine	C_5H_9NOS	Sulfur atom	*"Late" amino acids requiring specialized biosynthesis*
Methionine	$C_5H_{11}NO_2S$	Sulfur atom	
Phenylalanine	$C_9H_{11}NO_2$	Benzene ring	*Aromatic amino acids: need C=C ring formation*
Tyrosine	$C_9H_{11}NO_3$	Phenol ring	
Tryptophan	$C_{11}H_{12}N_2O_2$	Indole ring	
Arginine	$C_6H_{14}N_4O_2$	4 amine (C–NH_2) bonds	*Imines (C=N) dominate amines*
Lysine	$C_6H_{14}N_2O_2$	2 amine bonds	

These are precisely the "late" amino acids in evolutionary phylogenetics — incorporated into the genetic code only after more complex biosynthetic pathways evolved. Our simulation independently identifies the same early/late partition.

3.5 Beyond Amino Acids: Life's Complete Molecular Toolkit

A comprehensive scan of the Reactor900 log (2,000 steps, 5,056 unique formulas) reveals that the system discovers far more than amino acids. The full inventory of biologically significant molecules is striking:

3.5.1 All Five Nucleobases

The letters of RNA and DNA — the information carriers of all life — emerge spontaneously:

Nucleobase	Formula	First Step	Events	Role
Uracil	$C_4H_4N_2O_2$	**14**	22	RNA base
Thymine	$C_5H_6N_2O_2$	**46**	33	DNA base
Adenine	$C_5H_5N_5$	**127**	150	RNA + DNA base
Cytosine	$C_4H_5N_3O$	**204**	12	RNA + DNA base
Guanine	$C_5H_5N_5O$	**1,418**	4	RNA + DNA base

All five nucleobases emerge from bare atoms. Uracil appears at step 14 — earlier than most amino acids. Adenine (150 events) is abundantly produced. The emergence order (U → T → A → C → G) is consistent with the "RNA world" hypothesis, where uracil-based RNA preceded DNA.

3.5.2 The Formose Sugar Chain

The complete pathway from formaldehyde to glucose — the formose reaction central to prebiotic sugar synthesis — emerges step by step:

Sugar	Formula	First Step	Events	Carbons
Formaldehyde	CH_2O	1	36	1
Glyceraldehyde	$C_3H_6O_3$	218	94	3
Erythrose	$C_4H_8O_4$	1,114	4	4
Ribose	$C_5H_{10}O_5$	2,250	2	5
Glucose	$C_6H_{12}O_6$	2,527	1	6

Ribose ($C_5H_{10}O_5$) is the sugar backbone of RNA. Its emergence at step 2,250, together with all five nucleobases, means the system independently discovers the molecular components of RNA. Glucose appears at step 2,527 — the fuel molecule of all cellular life.

3.5.3 Krebs Cycle Intermediates

Several molecules from the citric acid cycle — the central metabolic pathway of all aerobic life — appear spontaneously:

Molecule	Formula	First Step	Events	Metabolic Role
Pyruvic acid	$C_3H_4O_3$	500	64	Glycolysis end product, Krebs entry
Succinic acid	$C_4H_6O_4$	127	2	Krebs cycle intermediate
Malic acid	$C_4H_6O_5$	1,550	8	Krebs cycle intermediate
Formic acid	CH_2O_2	928	23	C1 metabolism
Oxalic acid	$C_2H_2O_4$	2,042	6	Plant metabolism

3.5.4 Nucleotide Precursors and Peptide Chains

Molecule	Formula	First Step	Events	Significance
Imidazole	$C_3H_4N_2$	**7**	**432**	Core ring of histidine and purines
Pyrimidine	$C_4H_4N_2$	14	191	Core ring of C, T, U nucleobases
Aminopyrimidine	$C_4H_5N_3$	34	287	Cytosine precursor
Diketopiperazine	$C_4H_6N_2O_2$	**7**	31	Cyclic dipeptide (proto-protein)
Gly-Gly-Gly	$C_6H_{11}N_3O_4$	335	62	First tripeptide
Urea	CH_4N_2O	2,385	4	Nitrogen cycle, Miller-Urey product

The big picture. From 900 bare atoms, the system discovers the complete molecular toolkit for the origin of life:

- **Proteins:** 12 amino acids + dipeptides + tripeptide + cyclic peptide (diketopiperazine)
- **Genetic information:** all 5 nucleobases + imidazole/pyrimidine ring precursors
- **Energy metabolism:** pyruvate, succinate, malate (Krebs cycle intermediates)
- **Structural scaffolds:** formaldehyde → glyceraldehyde → erythrose → ribose → glucose

These are not cherry-picked targets — they emerge simultaneously from the same blind thermodynamic simulation. The prebiotic soup was not a random mess; it was a **thermodynamically inevitable precursor kit** for life.

3.6 Reactor Scaling: Four Sizes Overnight

Reactor	Atoms	Steps	Species	Total Goods	Largest Molecule	Entropy	AAs
R900	900	2,000	1,308	151K	$C_{48}H_{85}N_{13}$ (146 atoms)	4.96	**12/12**
R1800	1,800	2,000	1,301	160K	$C_{39}H_{56}N_{22}$ (117 atoms)	5.47	8/12
R3600	3,600	1,000	1,011	441K	$C_{104}H_{209}N$ (314 atoms)	3.43	**12/12**
R7200	7,200	500	790	152K	$C_{167}H_{336}$ (503 atoms)	4.34	10/12

Superlinear scaling: R3600 (4× atoms) produces ~8× the goods per step — an economy of scale from denser trade networks. **Entropy decreases** over time (5.67 → 4.96 bits), indicating the system matures from diverse exploration to concentration on thermodynamically favored products — the “winnowing” of prebiotic chemistry.

4. Discussion

4.1 Comparison with Existing Methods

Method	Time	Cost	Finds Pathways?	Finds Early 10?
Miller-Urey experiment	Days–weeks	Lab equipment	No (only products)	Yes
DFT / ab initio	Weeks per pathway	Supercomputer	1 at a time	Impractical
Molecular dynamics	Months	Supercomputer	Partially	Not yet achieved
Thermodynamic tables	Hours	Pen & paper	No (ranks only)	Ranks, not discovers
D-ABM (this work)	**5 minutes**	**Laptop**	**Yes (up to 260)**	**12/12**

4.2 Why Does the Free Market Work?

Parallel exploration 900 agents simultaneously explore different regions of chemical space, sharing discoveries through trade networks. This is massively parallel combinatorial search without any central coordinator.	**Demand cascades** When a complex molecule is demanded, the supply chain automatically pulls all intermediates — backward chaining from products to available reactants. The market "figures out" multi-step synthesis.
Thermodynamic selection Only exothermic reactions ($\Delta E < 0$) succeed, naturally pruning the vast combinatorial space. The market searches the chemical energy landscape the way evolution searches fitness landscapes.	**Emergent specialization** Workers self-organize into supply chains: some produce methane, others ethane, others combine them into larger molecules. No central planning — just local interactions and energy minimization.

4.3 The SMILES Caveat

Important limitation: The molecules produced have the correct **atomic formulas** but not necessarily the correct **structural isomers**. Our "glycine" ($C_2H_5NO_2$) has SMILES `N=CCOO` (an imine-peroxide), not `NCC(=O)O` (the classical α-amino acid). On the other hand, many smaller molecules emerge with the correct structure: formaldehyde appears as `C=O`, acetaldehyde as `CC=O`, and formic acid as `O=CO`.

This occurs because C=N (615 kJ/mol) is thermodynamically favored over C–N (305 kJ/mol). In prebiotic chemistry, tautomerization and environment-dependent rearrangements would interconvert between isomers. Our model captures the **thermodynamic accessibility of the formula**, which is the relevant quantity for predicting prebiotic amino acid abundance.

4.4 Thermodynamic Inevitability

Our results support the hypothesis that amino acids are **chemical inevitabilities**, not biological inventions. The 10 "early" amino acids found by our simulation match those identified independently by:

•	**1953**	— Miller-Urey spark discharge experiments
•	**1969**	— Murchison meteorite amino acid analysis
•	**2000**	— Phylogenetic reconstruction of the genetic code (Trifonov)
•	**2009**	— Thermodynamic free energy rankings (Higgs & Pudritz)
•	**2026**	— Free-market agent-based model (this work)

Five completely independent lines of evidence converge on the same set. The genetic code's amino acid alphabet was dictated by **thermodynamics**, not evolutionary contingency. Nature chose these amino acids because they were the most readily available — the thermodynamic winners of prebiotic chemistry.

5. Future Directions

Expand the chemistry	Beyond amino acids
• **Sulfur:** Add S atoms → cysteine, methionine • **Aromatic rings:** C=C in ring closure → Phe, Tyr, Trp • **Amine bonds:** Kinetic trapping of $C{-}NH_2$ → Arg, Lys	• **Peptide bonds:** Amino acid condensation into oligopeptides • **Weak interactions:** Van der Waals, H-bonds → micelles, membranes • **Compartmentalization:** Proto-cellular enclosure • **Autocatalysis:** Self-replicating reaction networks — the threshold of life

6. Conclusions

We have built a computational "time machine" — a time-lapse movie of the interior of a virtual micelle — that replays the first chapter of molecular evolution. Starting from nothing but atoms and thermodynamic rules, a free-market agent-based model spontaneously initiates the discovery of **life's complete molecular toolkit**: 12 amino acids, all 5 nucleobases, the formose sugar chain (formaldehyde → ribose → glucose), Krebs cycle intermediates, nucleotide ring precursors, and the first tripeptide.

The approach is **4–6 orders of magnitude cheaper** than existing computational methods and provides not just products but **complete synthesis networks** with multiple independent pathways. Beyond prebiotic research, the same framework could serve as a rapid exploration tool for **novel synthesis pathway design** in modern chemistry.

The result suggests that the emergence of life's molecular toolkit was not a matter of chance but of **thermodynamic necessity** — and that simple ***darwinian* economic *optimization*** principles (supply, demand, competition) may capture the essential dynamics of prebiotic chemical evolution.

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AI Assistance Statement

The preparation of this paper was significantly assisted by an AI coding assistant (Anthropic's Claude), which contributed implementation of the author's driving ideas and debugging of the simulation code (MarketSim engine, UFF evaluator, RDKit integration, timeout-protected threading), automated launching, monitoring, and result extraction from multi-reactor overnight batch runs, statistical analysis of amino acid emergence timing and pathway enumeration, literature surveys on prebiotic chemistry and thermodynamic amino acid rankings, and drafting of the manuscript — a domain (prebiotic chemistry and origin-of-life research) largely unfamiliar to the author, whose background is in semiconductor process modeling and agent-based economic simulation. This experience reinforces the observation from previous work (Jaraiz, 2026) that AI-assisted research may substantially lower the barriers for cross-disciplinary contributions, enabling domain outsiders to engage productively with fields where they lack formal training but can bring fresh methodological perspectives.

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